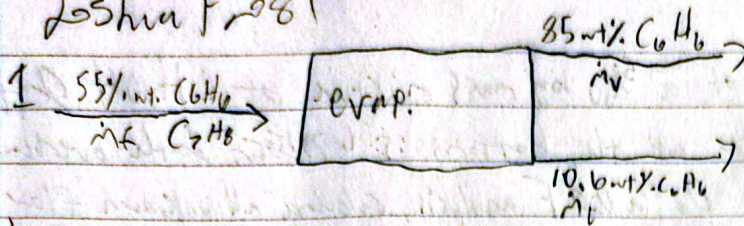


ChE 210 HW 3

Joshua Frost

2/6/26



$$b.) \dot{m}_F X_{OF} = \dot{m}_V X_{OV} + \dot{m}_L X_{OL} \rightarrow (\dot{m}_V + \dot{m}_L) X_{OF} = \dot{m}_V X_{OV} + \dot{m}_L X_{OL}$$

$$\rightarrow \frac{\dot{m}_V}{\dot{m}_L} = \frac{X_{OF} - X_{OL}}{X_{OV} - X_{OF}} = \frac{0.55 - 0.106}{0.65 - 0.55} = \boxed{1.48}$$

c.) DOF

$$(4 + 0) - (0 + 1 + 0) = 3 > 0 \rightarrow \text{underspecified}$$

P2.39.7 $1000 \text{ lb}_m \text{ NH}_3 / \text{hr} \times \frac{1 \text{ lb}_m \text{ NH}_3}{2 \text{ lb}_m \text{ NH}_3} = 500 \text{ lb}_m \text{ NH}_3 / \text{hr} / 0.25 = 2000 \text{ lb}_m \text{ NH}_3 / \text{hr}$

$1:3 = 2000 \text{ lb}_m \text{ NH}_3 / \text{hr} \times 3 = 6000 \text{ lb}_m \text{ H}_2 / \text{hr} = 8000 \text{ lb}_m \text{ H}_2 / \text{hr}$

$\rightarrow \dot{V} = \frac{\dot{n}RT}{P} = \frac{8000 \times 0.73024 \times 1410}{100} = \boxed{80986.36}$

2.54.2 a.) First feed = F, Superheated = S, Ferrous N = N

$\rightarrow \text{Glucose: } 0.45 \text{ M}_F = 0.15, \text{ Phos: } 0.02 \text{ M}_F = 0.35 \text{ M}_S + 0.12 \text{ M}_N = 0.06$

0.45	0.02	0.01	55	0.325	$\rightarrow X_{OF}$	3250 F
0	0.35	0.18	5	0.139		1395 S
0	0.12	0.58	N	0.075		750 N

b.) $20(0.325) + 10(0.139) + 15(0.075) = 9.02 / \text{kg}$

$\rightarrow 15 - 9.02 = \text{save } 5.98 / \text{kg}$

c.) $0.325(0.08) + 0.139(0.08) = 0.0209 = 2.09 \text{ wt\% H}_2\text{O}$

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explanation? 6 pts

2.60.7 $5.6 \text{ Eq} \times 2000 \text{ lbs} / \text{ton} / 3 \text{ Eq/ton} = 3733.3 \text{ lbs/person}$

$\rightarrow$ This does not seem right, & isn't because it's taking the total emissions & putting that safety on people & excluding large manufacturing plants, corporations & celebrities with excessive private jet usage.

add 1.0 pts

5/15

210 H₂O 3 cont.

2.64.) ~~DOF~~

$F_w + F_A + F_B = 200 \text{ kg/day}$

$H_2SO_4: F_A(0.77) + F_B(0.108) = 200(0.186) = 37.2$ (costs)

DOF: $3 - 3 = 0 \Rightarrow$ well specified

$\rightarrow F_B = \frac{37.2 - 0.77 F_A}{0.108} = 344.4 - 7.13 F_A$

$C = 10(F_A) + 68888 - 14.26 F_A = 68888 - 4.26 F_A$

$0.77(F_A) + 0.108(344.4 - 7.13 F_A) = 37.2 \rightarrow F_A = \frac{37.2}{0.77} = 48.312 \text{ kg/day}$

$48.31 + 0 + F_w = 200 \rightarrow F_w = 151.69 \text{ kg/day}$

It is more cost eff. to just use F_w & F_A

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